

Dorsal hippocampus mediates light-tone sensory preconditioning task in mice

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eLife Assessment

Pinho et al use *in vivo* calcium imaging and chemogenetic approaches to examine the involvement of hippocampal sub-regions across the different stages of a sensory preconditioning task in mice. They find **convincing** evidence for sensory preconditioning in male mice. They also find that, in these mice, CaMKII-positive neurons in the dorsal hippocampus: (1) encode the audio-visual association that forms in stage 1 of the task, and (2) retrieve/express sensory preconditioned fear to the auditory stimulus at test. These findings are supported by evidence that ranges from **incomplete** to **convincing**. The study will be **valuable** to researchers in the field of learning and memory.

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Abstract

Daily choices are often influenced by environmental cues that are not directly linked to reinforcers. This process is known as higher-order conditioning and can be measured using sensory preconditioning tasks in rodents. This behavioral paradigm requires the repeated and simultaneous presentation of two low-salience stimuli, such as a light and a tone, followed by a devaluation phase where one stimulus is paired with an unconditioned stimulus, such as a mild footshock. The outcome is a conditioned response (i.e. freezing response) to both the conditioned stimulus (direct learning) and the non-conditioned stimulus (mediated learning). In our study, we set up a successful light-tone sensory preconditioning task in male and female mice. Sex differences were seen on the number of conditioning sessions required to acquire mediated learning and in the behavioral responses observed in certain control experimental groups. We used *in vivo* calcium imaging to characterize the activity of hippocampal neurons in the dorsal and ventral subregions of the hippocampus when associations between low-salience stimuli and reinforcers occur. Finally, we combined our sensory preconditioning task with chemogenetic approaches to assess the role of these two hippocampal subregions in mediated learning. Our results indicate that dorsal, but not ventral, CaMKII-positive cells mediate the encoding of low-salience stimuli during the preconditioning phase. Overall, we implemented a novel light-tone sensory

preconditioning protocol in mice that allowed us to detect sex differences and to further elucidate the role of particular hippocampal subregions and cell types in regulating these complex cognitive processes.

Introduction

The brain mechanisms underlying associative learning have been classically elucidated through classical conditioning paradigms using strong salient stimuli as reinforcers such as an electric footshock paired with low-salience stimuli (e.g., light, tone, odor, etc). These important cognitive processes are not sufficient to fully understand animal behavior because daily choices are not always dictated by stimuli that have been directly associated with a potent reinforcer. Indeed, humans and animals frequently respond to cues that were never explicitly conditioned, but they happened to be previously associated with other stimuli paired with a specific aversive or rewarding meaning^{1,2,3,4,5}. These higher-order conditioning processes, also known as mediated learning, can be captured in laboratory settings through sensory preconditioning procedures^{2,3,6,7,8,9,10,11}. These tasks combine incidental associations between low-salience stimuli (e.g., odors, tastes, lights or tones) during a preconditioning phase followed by a classical conditioning of one of these stimuli with an aversive or appetitive unconditioned reinforcer^{2,3,6,7,8,9,10,11}. As a result of these processes, subjects present aversion or preference to the stimulus never explicitly paired with the reinforcer, therefore allowing the evaluation of mediated learning^{2,3,6,7,8,9,10,11}. Whereas brain circuits underlying classical direct associative memory between neutral cues and reinforcers have been largely studied, the brain mechanisms underlying incidental associations between low-salience sensory stimuli and their impact on behavior have been much less explored.

Hippocampus and other cortical regions (e.g., perirhinal, retrosplenial and orbitofrontal cortices) are involved in higher-order conditioned responses^{6,7,8,9,10,11,12,13,14} assessed through different sensory preconditioning tasks using different sensory modalities. Previous findings suggest that the involvement of several brain regions in higher-order conditioning might depend on the different behavioral phases studied (i.e., preconditioning or testing) or the different sensory modalities used (visual, gustatory, olfactory or auditive cues). However, it is still unknown if there could be common brain locations where the different stimuli are integrated independently of the behavioral protocol or sensory modality used. In this sense, the hippocampus has been suggested to play an important role in different behavioral phases of sensory preconditioning procedures across species^{1,2,3,4,5,11}. This brain region, which continuously exchanges information with other cortical areas, has been proposed as a possible integrator of previous experiences into several representations through a precise excitatory-inhibitory balance^{15,16}. Indeed, previous studies pointed to a specific role of hippocampal mechanisms during the encoding of incidental associations¹¹. However, the role of different hippocampal subregions or cell types in the formation of mediated learning is still not understood. Indeed, the dorso-ventral axis of the rodent hippocampus is structurally and functionally segregated^{17,18}. The dorsal hippocampus, which is connected with cortical regions and the thalamus, is mainly involved in cognitive functions such as navigation and exploration^{17,18,19,20,21}. On the other hand, the ventral hippocampus is mainly connected to the amygdala, nucleus accumbens and hypothalamus, and is involved in motivated and emotional behavior^{17,18,19,20,22}. In this context, our initial hypothesis was that dorsal and/or ventral hippocampus could be differentially involved in mouse sensory preconditioning.

We took advantage of chemogenetic and imaging approaches using an adapted light-tone sensory preconditioning⁶ (L_TSPC) procedure in mice, to suggest that the activity of dorsal hippocampus, in particular the activity of CaMKII-positive cells, is crucial for the formation of incidental associations between lights and tones, eventually leading to mediated learning. Understanding the

differential contribution of dorsal and ventral hippocampus to sensory preconditioning tasks provide valuable insights into the mechanisms underlying higher-order conditioning and the broader functioning of the hippocampus in associative processes.

Material and methods

Animals

Male and female C57BL/6J mice were purchased from Charles River Laboratories and were used for behavioral and chemogenetic experiments. Male Parvalbumin (PV)-Cre mice were obtained by own breedings in our animal facility. All the mice used in this study were 8 weeks old at the beginning of the experiments and they were grouped-housed and maintained in a temperature (20–24°C) and humidity (40%–70%) controlled condition under a 12 h light/dark cycle and had ad libitum access to food and water. All behavioral studies were performed during the dark cycle (from 9am to 5pm) by trained researchers who were blind to the different experimental conditions.

All experimental procedures shown in this work have followed European guidelines (2010/62/EU) and were approved by the Committee on Animal Health and Care of Barcelona Biomedical Research Park (PRBB) and from the Generalitat de Catalunya. All experiments were performed in the animal facility of the PRBB, which has the full accreditation from the Association for Assessment and Accreditation of Laboratory Animal Care (AAALAC).

Light-tone sensory preconditioning task ($_{LT}$ SPC)

The task was performed using automatized conditioning chambers (Imetronic, France) and was divided in different phases (**Figure 1A** [↗](#)):

Habituation

Animals were exposed to the chambers with the presence of background noise (65db) (similar to white noise but allocated to different speakers to avoid that animals associate the sound with a location) during one session of 20 minutes.

Preconditioning Phase

This phase was composed by six sessions (2 per day; 3 hours of intersession interval) of 510 seconds each under background noise. These sessions started with an OFF period of 3 min and were followed by 5 simultaneous presentations of light (CS1, white LED light located in one side of the box) and tone (CS2, 65db, 3000Hz) during 30 seconds with an intertrial interval of 30 seconds and a final OFF period of 1 minute. This protocol details apply to the Paired and No-Shock experimental groups. For the unpaired group, light and tone were never associated together during this phase. Indeed, animals were exposed to the same amount of time to light and tone but in different sessions (S1 light, S2 tone, S3 tone, S4 light, S5 light, S6 tone, in a pseudo-randomized way).

Conditioning Phase

This phase consisted of two training sessions in males and one training session in females. Each session last 510 seconds under background noise. In case of males, there was an inter-session interval of 3 hours. Specifically, the sessions started with an off period of 3 min and were followed by 5 presentations of a 10-seconds light stimulus (CS1) that co-terminate during 2 seconds with a mild footshock (0.4 mA, see results) with an intertrial interval of 1 min and a final OFF period of 1

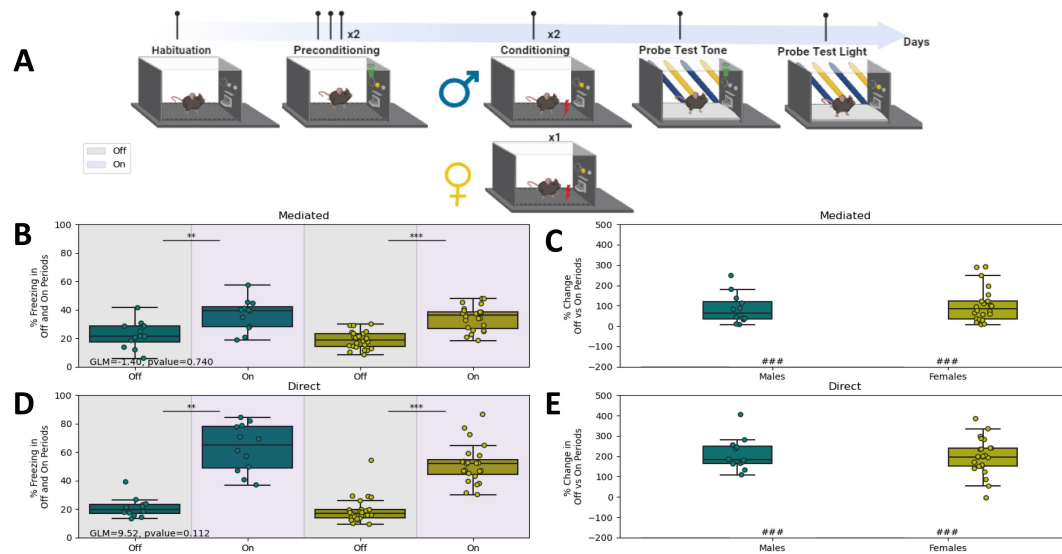


Figure 1.

Male and female mice performed auditory-visual induced mediated learning.

A) Schematic representation of the L_T SPC task highlighting the experimental differences in male (2 training sessions during conditioning) and female (1 training session) mice. The percentage of time spent freezing during OFF and ON periods of the probe test 1 (Tone) (B) and probe test 2 (light) (D). The percentage of change of the OFF vs ON periods of the probe test 1 (Tone) (C) and probe test 2 (light) (E). GLM: generalized linear model fitted to gamma distribution with planned comparisons (B, D), * significant p-value (<0.05) after false discovery rate (FDR). Mann-Whitney between sexes (C, E) and # significant against 0 after tested by Wilcoxon test with zero method. See statistical details in Supplementary Table 1.

min. This protocol details apply to the Paired and Unpaired experimental groups. For the No-Shock group, animals followed the same conditioning session but without the exposure to the electric footshock.

Probe Tests Phase

Mice were subjected to two probe test sessions on the same day where they were exposed to the tone (Mediated Learning, Probe Test 1) and the light (Direct Learning, Probe Test 2) in a new context (different from the previous phases). In particular, we changed 4 features of the context: floor texture, wall colors, smell (from 70% ethanol to a CR36 disinfectant solution), and absence of background noise to avoid fear responses elicited by the context itself. The two Probe Tests lasted 6 minutes and were separated for at least 1 hour. In this session, animals underwent an OFF period of 3 min followed by an ON period of 3 min where the Tone (Probe Test 1) or the Light (Probe Test 2) were continuously exposed. All experimental groups (Paired, Unpaired and No-Shock) perform the Probe Tests in the same manner. The presence of mediated and direct learning is shown by the freezing time duration during ON and OFF periods or with the percentage of change between these periods. This percentage of change of each individual comparing OFF and ON periods were calculated using the formula:

$$\%change = \frac{(time\ freezing\ during\ ON\ period - time\ freezing\ during\ OFF\ period)}{time\ freezing\ during\ OFF\ period} \times 100$$

The behavior analysis was automatized using Deeplabcut to track the animal's position across time and a Python homemade script to compute all the different analyses, graphs, and statistics performed in the present paper. The main behavior measured was freezing response (i.e. immobility), which was defined as an euclidean distance lower than 0.02 cm per pair of frames for videos with 25 fps, resulting in a speed lower than 0.5 cm/s. We validated this automated behavioral counting by performing correlations with manual counts and another software dedicated to freezing scoring (EzTrack)²³, in which we found a high correlation (higher than 90%) (Supplementary Figure 1). The Deeplabcut project and all scripts will be available in the GitHub repository of the lab.

In vivo Fiber Photometry

Male PV-Cre mice were anesthetized with a mixture of ketamine (75 mg/Kg, Imalgene 500, Merial, Spain) and medetomidine (1 mg/Kg, Domtor, Spain) by intraperitoneal (i.p.) injection. Then, animals were placed into a stereotaxic apparatus (World Precision Instruments, FL, USA) with a mouse adaptor and lateral ear bars. For viral intra-hippocampus delivery, AAV vectors were injected with a Hamilton syringe coupled with a nanofill attached to a pump (UMP3-1, World Precision Instruments, FL, USA). Mice were injected with a mixture of 200 nl of AAV.Syn.NES.jRCaMP1a.WPRE.SV40 (titer: 1×10^{13} , addgene 100848-AAV9) and 200nl of a Cre-dependent AAV-syn-FLEX-jRCaMP8f-WPRE (titer: 1×10^{13} , addgene 162379-AAV9) in order to monitor both the activity of the synapsin-positive cells in red and the PV interneurons in green. These viral vectors were infused (1 nl/s) directly into the dorsal (in one hemisphere) or ventral hippocampus (in the other hemisphere) with the following coordinates in mm: dorsal, AP ± 1.5 , ML ± 2 , DV -1.5 and ventral, AP ± 3.5 , ML ± 3.3 , DV -3.5 , according to Paxinos and Franklin brain atlas (Paxinos and Franklin, 2001). After the AAV infusions, an optic Fiber (core 400 μ m, N.A 0.5, RWD, China) was implanted using dental cement following the same coordinates except for DV that was placed 0.25 mm above viral infusions. Four weeks after this surgery, animals were used for *in vivo* calcium recordings where they undergo a LT SPC and *in vivo* recordings were performed during Light-Tone associations (preconditioning phase) and Light-Shock associations (conditioning phase). Before this experiment, animals were habituated for 3 days to connect and disconnect the optic fibers and to be habituated to the cable in the same chamber where the behavioral experiment was performed. During these two behavioral phases, *in vivo* recordings were

performed using a commercial Fiber Photometry equipment (RWD, China) where we used 470nm LED to excite the GCaMP sensor, and 560 nm for the RCaMP signal. In all mice used, after the recording experiments, the signal was checked to validate the viral infusions.

To analyze the Fiber Photometry experiments, a custom python code was used and the behavioral videos and photometry recordings were synchronised by TTL signals. In particular, raw calcium signals were pre-processed by removing the first minute of the recording to decrease the effect of the initial exponential photobleaching, and by removing point artifacts. The 470nm signal was fitted to the isosbestic 405nm using a linear fit and for each time point, $\Delta F/F$ was calculated as $(F_{470nm} - F_{405nm(fitted)})/F_{405nm(fitted)}$. This procedure is the same as described in previous works²⁴. The codes used for the analysis will be found on the dedicated github repository.

Chemogenetic modulation

Stereotaxic surgeries were performed as described above for fiber photometry. In this case, C57BL/6J mice were injected with 500 nl AAV-CaMKII-mCherry (titer: 7×10^{12} , 114469-AAV5) and 500 nl AAV-CaMKII-hM4Di (titer: 7×10^{12} , 50477-AAV2) directly into the hippocampus, with the following coordinates: dorsal, AP ± 1.5 , ML ± 2 , DV -1.5 and ventral, AP ± 3.5 , ML ± 3.3 , DV -3.5 , according to Paxinos and Franklin brain atlas (Paxinos and Franklin, 2001). On the other hand, PV-Cre mice were infused with AAV-DIO-hM4Di using the same coordinates to target the dorsal hippocampus. Three control groups were performed for each subregion (dorsal and ventral hippocampus): animals infused with control virus (pAAV-CaMKIIa-mCherry) and injected with saline or JHU37160 dihydrochloride (J60, 0.1 mg/Kg, i.p., HelloBio, HB6261)²⁵, and animals infused with AAV-CaMKII-mCherry or AAV-DIO-hM4Di injected with saline. Animals were used for chemogenetics experiments four weeks after injections to get an optimal expression of the viruses. The injection of J60 was one hour before each preconditioning session (C57BL/6J and PV-Cre mice) or before the Probe Test 1 (C57BL/6J mice). In all mice used in the behavioral experiments, the signal was checked, and representative images are shown in Supplementary Figures.

Histology

After the chemogenetic and imaging experiments, mice were anesthetized i.p with a mixture of ketamine (50 mg/Kg) and xylazine (20 mg/Kg) in overdose (3-4x body weight), transcardially perfused with cold 4% paraformaldehyde to fix tissues. Brains were removed and sectioned in serial coronal sections of 20 μ m and collected directly to the slide for further analysis in the microscope. All sections were counterstained with 4',6-diamidino-2-phenylindole (DAPI, ref. 00-4959-52, Fluoromount-G w/DAPI, Life Technologies, USA) to visualize cell nuclei in the mounting medium. Slides were coverslipped and imaged by an epifluorescence Nikon Eclipse Ni-E. All animals were signal checked to guarantee the target of the hippocampus subregion.

Data collection and statistical analysis

Data collection

All mice were randomly assigned to experimental conditions. Researchers performing the experiments were always blind to the condition of the subject and we used an automated way to analyze behavioral responses throughout the study. Raw data was processed and analyzed using homemade python packages.

Statistical analysis

Graphs and statistical analysis were performed through homemade python scripts that will be openly shared. All data comes from distinct mice and is shown as independent data points per animal $\pm$ standard error of the median (SEM). Normality and homoscedasticity of the data were assessed with the Kolmogorov-Smirnoff and Levene tests, respectively. Due to non-parametric

properties of the data, a generalized linear model with gamma distribution was used for multivariable analysis (after a goodness of fit to select the appropriate distribution), followed by planned comparisons corrected with false discovery rate. For univariate analysis, the Kruskal-Wallis test were performed followed by planned comparisons corrected with false discovery rate. For simple comparisons the Mann-Whitney test were performed, and for comparisons of indexes against zero the Wilcoxon test with zero method were used. Detailed statistical data for each experiment can be found in Extended Table 1 (for Main Figures) and Extended Table 2 (for Supplementary Figures).

Results

Female and male mice show mediated learning using an auditory-visual sensory preconditioning task

Sensory preconditioning tasks using olfactory and gustatory stimuli were already demonstrated in mice in several previous works^{11,14}. However, other sensory modalities such as visual and auditory stimuli are also very relevant for animal daily choices. Whereas the use of these cues to assess aversive sensory preconditioning was also shown in rats^{6,8,13}, the use of auditory and visual cues in mice is limited¹¹, especially for aversive paradigms. The first aim of the present work was to set up a mouse sensory preconditioning task using auditory and visual cues to be able to assess mediated and direct learning in mice. In this regard, a sensory preconditioning protocol was established (see Methods, **Figure 1A**) together with the use of a novel automatized tool to analyze freezing response (i.e. immobility) in mice by combining DeepLabCut pose estimation and a new generated python script. This novel tool was validated and highly correlated with manual counting and the ezTrack²³ freezing scorings (Supplementary Figure 1).

Using male and female mice, we observed that the exposure to the Tone (Probe Test 1, ON period) induced an increased freezing time (**Figure 1B**) in relation to its absence (OFF period) and also a significant increase in the percentage of change (**Figure 1C**), suggesting the presence of mediated learning. When the light was presented (Probe Test 2), animals showed an increase in freezing response in the ON period in comparison with the OFF period (**Figure 1D**) and increased percentage of change in the ON period (**Figure 1E**), revealing the existence of direct learning. Both measures, percentage of time in freezing and percentage of change, revealed no significant changes between male and female mice. However, male mice needed to be trained with two conditioning sessions, whereas female mice were trained only with one session, to see a comparable mediated and direct learning in both sexes. During the set-up of this paradigm (Supplementary Figure 2), we realized that males with 1 training session presented lower expression of mediated learning in comparison with males that performed 2 training sessions or females with 1 training session. This data suggests potential sex-dependent differences in the sensory preconditioning task, where each protocol need to be adapted depending on the sex used.

Simultaneous light-tone associations during preconditioning are required for mediated learning formation

At this point, we wanted to discard any bias due to direct behavioral effects induced by the exposition to certain sensory cues in order to fully validate our _{LT}SPC task. We performed a _{LT}SPC experiment with three different experimental groups (**Figure 2A**, see Methods): Paired (exact protocol conditions as shown in **Figure 1**), Unpaired (same protocol as the paired group but light and tone were never associated together) and No-Shock groups (same protocol as the paired group but without the exposure to the electric footshock). Following this experimental design, male (**Figure 2B** and **2D**) and female (**Figure 2C** and **2E**) mice from the paired group showed mediated and direct learning, respectively. In particular, a higher amount of time exhibiting freezing was observed during the ON period when compared with the OFF period and

the percentage of change was significantly different from zero. These results clearly reproduced the data shown above and confirmed the existence of mediated and direct learning in male and female mice. However, when light and tone were separated on time (Unpaired group), male mice were not able to exhibit mediated learning response (**Figure 2B**) whereas their response to the light (direct learning) was not affected (**Figure 2D**). On the other hand, female mice still present a lower but significant mediated learning response (**Figure 2C**) and a normal direct learning (**Figure 2E**). These results identified another sex difference between female and male mice regarding their performance in sensory preconditioning tasks. Finally, in the No-Shock group, both male (**Figure 2B** and **2D**) and female mice (**Figure 2C** and **2E**) did not present either mediated or direct learning, which also confirmed that the exposure to the tone or light during Probe Tests do not elicit any behavioral change by themselves as the presence of the electric footshock is required to obtain a reliable mediated and direct learning responses. Altogether, this data confirmed that we successfully set up a L_T SPC protocol in mice and that this behavioral paradigm can be used to further study the brain circuits involved in higher-order conditioning.

Hippocampal cells are engaged during L_T SPC

Previous findings with sensory preconditioning paradigms using other sensory modalities suggested a key role of the hippocampus during the processing of incidental associations between low-salience stimuli during the preconditioning phase^{1,11}. To better characterize the involvement of dorsal and ventral hippocampus during our L_T SPC task, we conducted a simultaneous *in vivo* fiber photometry recordings with an optic fiber in each hippocampal subregion to monitor the activity of neurons in general (RCaMP activity) and the PV-positive interneurons in particular (GCaMP activity) during the preconditioning phase (*i.e.* pairing between light and tone) and the conditioning phase (*i.e.* pairing between light and footshock). Thus, we infused a mixture of an AAV-Syn-RCaMP and AAV-DIO-GCaMP (see Methods) into the dorsal (dHPC) or ventral (vHPC) subregion of the hippocampus of PV-Cre mice to be able to monitor the activity of different hippocampal cells. Four weeks after surgeries, animals underwent the L_T SPC task (**Figure 3A**). Both the activity of dorsal and ventral hippocampal cells are measured by $\Delta F/F$ after the pairings in L_T SPC (dHPC, **Figure 3B**, left and vHPC, **Figure 3C**, left). Peri-event time histograms computed on the Z-scored $\Delta F/F$ show the average dynamics of activity in both hippocampal nuclei during light-tone associations (dHPC, **Figure 3B**, right and vHPC, **Figure 3C**, right). This quantifies, trial-by-trial, the response variability, which is lower in the dHPC, and its evolution over time, which is longer lasting also in the dHPC. Quantification of this increase is performed by measuring the peak value of the Z-scored $\Delta F/F$ in a 1-second window after the stimulus presentation and comparing it to the baseline value (**Figure 3D**). In addition, both activity markers in each hippocampal subregion clearly increase their activity during the conditioning sessions both at the onset of the light (*i.e.* conditioned stimulus) (Supplementary Figure 3) and when the footshock is applied (*i.e.* unconditioned stimulus) (Supplementary Figure 4). Altogether, these results demonstrate the engagement of hippocampal cells from both hippocampal subregions during the associative sessions (*i.e.* light-tone and light-shock pairings) in our L_T SPC task in mice.

The dorsal hippocampus mediates L_T SPC

The *in vivo* photometry results obtained suggest a major engagement of the dorsal hippocampal subregion with an increase on the activity of different hippocampal cells during the onset of light-tone presentations in the preconditioning phase. However, no study has explored so far, a potential causal distinct role between dorsal and ventral hippocampus in mouse sensory preconditioning. To test this idea, adeno-associated viral vectors carrying the expression of an inhibitory DREADD (hM4DGi, hereafter called DREADD-Gi)²⁶ under the CaMKII promoter were infused into the dHPC and vHPC of mice to be able to inhibit the activity of principal neurons from these two hippocampal sub-regions. Control animals injected with saline or the DREADD agonist

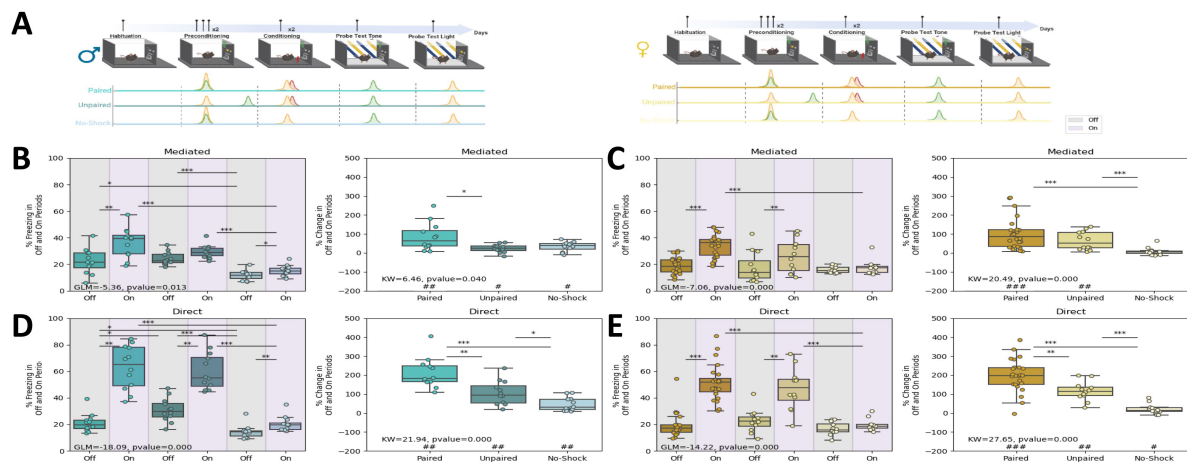


Figure 2.

Simultaneous associations between light and tone are required for mediated learning formation.

(A) Schematic representation of the L_T SPC task in males (2 training sessions) and females (1 training session) with a representation of paired, unpaired and no-shock experimental groups. The percentage of time spent freezing during OFF and ON periods of the probe test 1 (Tone) (B in males, C in females) and probe test 2 (light) (D in males, E in females). The percentage of change of the OFF vs ON periods of the probe test 1 (Tone) (B in males, C in females) and probe test 2 (light) (D in males, E in females). * significant p-value (<0.05) after false discovery rate (FDR). GLM: generalized linear model fitted to gamma distribution with planned comparisons. KW: Kruskal-Wallis across experimental groups: paired, unpaired, no-shock. # significant against 0 after Wilcoxon test with zero method. See statistical details in Supplementary Table 1.

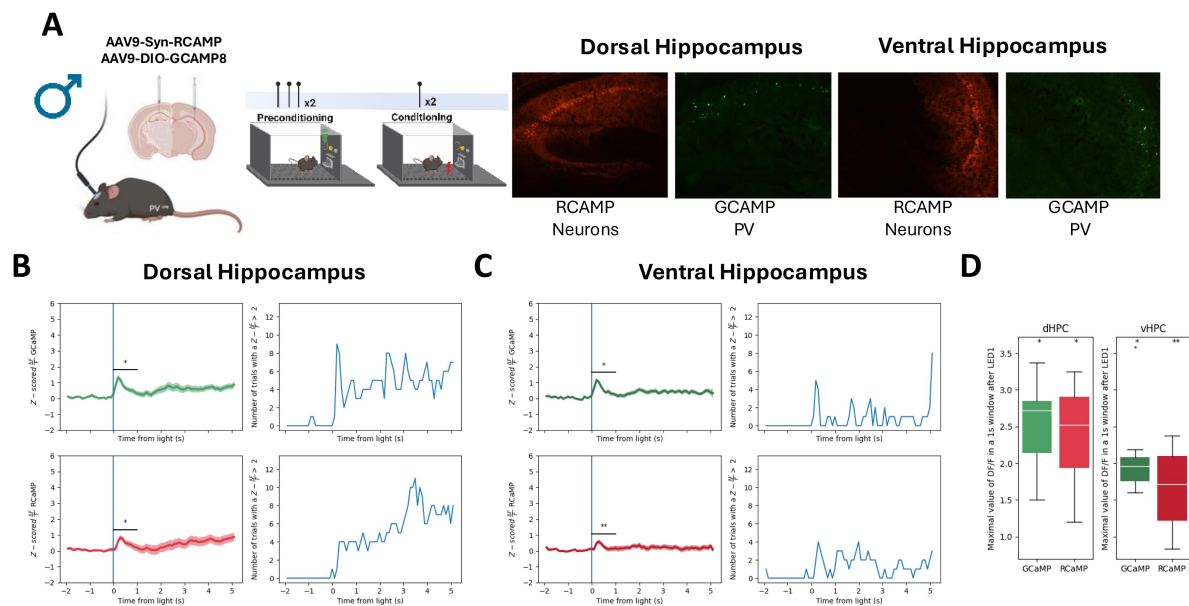


Figure 3.

Hippocampal cells during L_{TSPC} .

A) Schematic representation of L_{TSPC} task while fiber photometry recordings of RCaMP (calcium sensor in synapsin-positive neurons) and Cre-dependent GCaMP (calcium sensor in PV-positive interneurons) in dHPC and vHPC of PV^{Cre} mice. B) dHPC modulation during preconditioning of L_{TSPC} : on the left upper panel z scores of $\frac{\Delta f}{f}$ (where f represents fluorescence) of GCaMP PV-positive interneurons on dHPC (green), left bottom z scores of $\frac{\Delta f}{f}$ of RCaMP in neurons of dHPC (red), right upper number of events with a $\frac{\Delta f}{f} > 2$ of GCaMP PV-positive interneurons on dHPC, and right bottom number of trials with a $\frac{\Delta f}{f} > 2$ of RCaMP in neurons of dHPC. C) vHPC modulation during L_{TSPC} : on left upper panel z scores of $\frac{\Delta f}{f}$ of GCaMP PV-positive interneurons of vHPC (green), left bottom z scores of $\frac{\Delta f}{f}$ of RCaMP in neurons of vHPC (red), right upper number of trials with a $\frac{\Delta f}{f} > 2$ of GCaMP in PV-positive interneurons of vHPC, and right bottom number of trials with a $\frac{\Delta f}{f} > 2$ of RCaMP in neurons of vHPC. D) maximal value of $\frac{\Delta f}{f}$ in the first 1 second window after pairings compared with baseline, on the left (dHPC) and on the right (vHPC). * significant p-value (<0.05). See statistical details in Supplementary Table 1.

J60 show a reliable mediated and direct learning in the $_{LT}SPC$ task (**Figure 4**). Notably, the inhibition of CaMKII-positive neurons in the dHPC (*i.e.* J60 administration in DREADD-Gi mice) during preconditioning (**Figure 4B**), but not before the Probe Test 1 (**Figure 4B**), fully blocked mediated, but not direct learning (**Figure 4D**). In contrast, the inhibition of CaMKII-positive cells from the vHPC during preconditioning or Probe Test 1 did not impact mediated (**Figure 4C**) or direct (**Figure 4E**) learning. This data suggests that dHPC, but not vHPC, mediates the encoding of low-salience stimuli, such as light and tone, associations that are crucially involved in the formation of mediated learning. Once the dorsal hippocampus was identified as a key brain region modulating mediated learning, we wanted to investigate if the specific inhibition of PV interneurons, which were also engaged during stimuli presentation, was also impacting this particular behavior.

For that, we infused a Cre-dependent DREADD-Gi into the dHPC of PV-Cre mice. Notably, the inhibition of PV interneurons during light-tone associations was not impacting the mediated or direct learning formation (Supplementary Figure 5). Altogether, these results suggested that CaMKII-positive cells, but not PV interneurons, in the dHPC are key mediators of light-tone associations that lead to the formation of mediated learning.

Discussion

Our study provides novel insights into the involvement of the hippocampus in higher-order conditioning by validating a $_{LT}SPC$ task in male and female mice, using *in vivo* calcium recordings to link the activity of particular hippocampal cells from different hippocampal subregions to different behavioral phases of the task and, finally, using chemogenetic approaches to causally link the dHPC, but not the vHPC, with the encoding of light-tone pairings, which are required for the formation of mediated learning.

Sex differences in classical fear conditioning have been previously found with several contradictory results as stronger conditioning in males or females or no differences^{27–34} have been reported. However, previous experimental designs aimed at exploring sex differences undergoing sensory preconditioning paradigms have not been conducted. Our results indicate that both male and female mice can show mediated learning, but the protocol settings might differ between sexes as female mice require fewer conditioning sessions than male mice. This result could suggest that female mice are better in direct aversive conditioning by acquiring higher salience by its pairing with a reinforcer, which promotes mediated learning with less conditioning sessions than in male mice. Accordingly, previous works have shown significant differences in conditioned freezing responses between sexes^{35–38}.

When low salience stimuli were presented separated on time or when the electric footshock was absent, mediated and direct learning were abolished in male mice. In female mice, although light and tone were presented separately during the preconditioning phase, mediated learning was reduced but still present, which implies that female mice are still able to associate the two low-salience stimuli. No study has addressed sex differences in the capability of associating low-salience cues. However, although the salience of the pairing is different, it has been shown that female mice responded more than male mice to unpaired associations between a neutral stimulus and a reinforcer^{32,39}. As female mice normally exhibit more generalization of fear^{28,40,41}, it is plausible that the sex differences observed in the unpaired group, could be related to this enhanced fear generalization in females and not to a better ability of female mice to associate stimuli that are presented separately³².

Previous findings showed a crucial role of hippocampal cells in encoding location, context, cues, and memory expression associated with appetitive and aversive behaviors^{42–46}. However, few studies have explored the causal involvement of these hippocampal cells in mediating

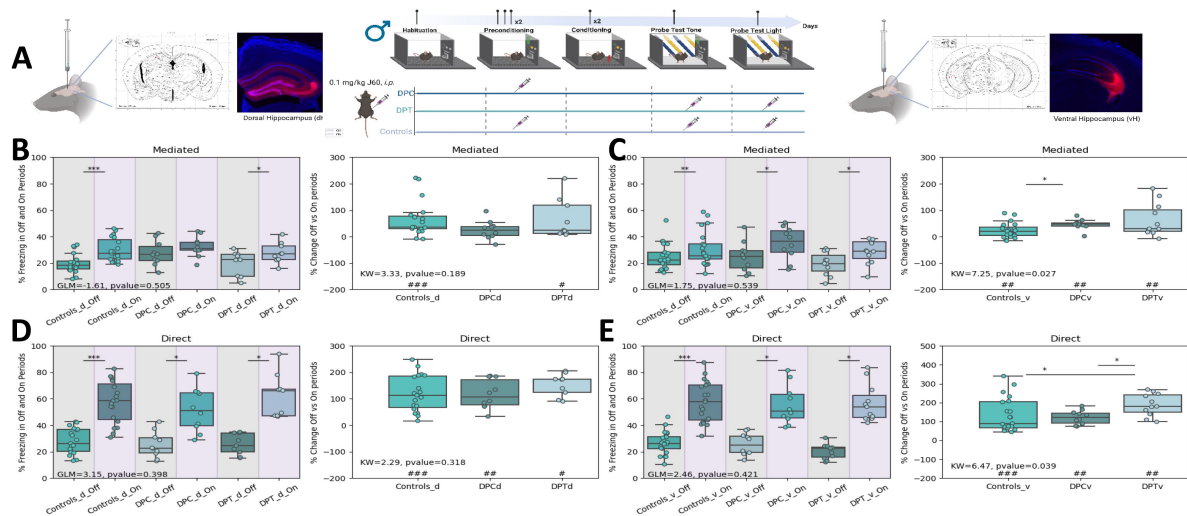


Figure 4.

Chemogenetic modulation of dorsal and ventral hippocampus during $_{LTSPC}$.

A) Schematic representation of $_{LTSPC}$ task combined with chemogenetic approaches, where an inhibitory DREADD was infused in dHPC (representative image on the left) and vHPC (representative image on the right). During the preconditioning (DPC) or the probe test (DPT), a DREADD agonist (J60) was injected intraperitoneally (controls where injected with the agonist in both phases) (diagram of DREADD agonist administration on the center). B, C) The percentage of time spent in freezing during OFF and ON periods (left) and the percentage of change (right) of the OFF vs ON periods of the Probe Test 1 (Tone) of animals infused in dHPC (B) and vHPC (C): controls (Controls_d_Off and controls_d_On, Controls_v_Off and controls_v_On), DPC (DPC_d_Off and DPC_d_On, DPC_v_Off and DPC_v_On) and DPT (DPT_d_Off and DPT_d_On, DPT_v_Off and DPT_v_On). D, E) The percentage of time spent in freezing during OFF and ON periods (left) and the percentage change (right) of the OFF vs ON periods of the probe test 2 (Light) of animals infused in dHPC (D) and vHPC (E): controls_d, DPC_d, DPT_d and controls_v, DPC_v, DPT_v). *significant p-value (<0.05) after false discovery rate (FDR). GLM: generalized linear model fitted to gamma distribution with planned comparisons (B, D, F, H). KW: Kruskal-Wallis across experimental groups: controls, DPC, DPT (C, E, G, I). # significant against 0 after Wilcoxon test with zero method. See statistical details in Supplementary Table 1.

reinforcement learning⁴⁷ and, indeed, their possible role in the encoding of the associations between low-salience stimuli have not been explored. In this study, we have used *in vivo* fiber photometry to elucidate a novel role of hippocampal cells from the dorsal and ventral hippocampus underlying the pairings between two low-salience stimuli such as light and tone and between one of these stimuli and an electric footshock. According to previous findings⁴⁷, we demonstrated that both the conditioned stimulus (*i.e.* light paired with the footshock) and the unconditioned stimulus (*i.e.* footshock) induced a large increase in the activity of hippocampal cells. The activation of particular neurons from the hippocampus are also engaged in reinforcement behavior^{48,49}. In contrast with all these findings, it has been also proposed that the hippocampus is not required for learning simple cue relationships such as associating a conditional stimulus (CS) with an unconditional stimulus (US), which is mediated by other neural circuits (*e.g.*, the amygdala)⁵⁰.

According to the data obtained by our photometry recordings, the simultaneous exposure to different stimuli is encoded by the hippocampus, being important for its role in associative learning and for integrating sensory information from the environment⁵¹. Thus, it is not surprising that previous studies supported the function of this brain region during the associative phases (*i.e.* preconditioning and conditioning phases) of sensory preconditioning tasks both in humans and in animals^{1,11,13,14,52}. However, none of these studies have explored the specific engagement of different hippocampal subregions or cell-types during these higher-order conditioning tasks. Indeed, our data suggests that when hippocampal activity is modulated by the specific manipulation of hippocampal subregions, this brain region is not involved during retrieval. In addition, the present results reveal that the dHPC, but not the vHPC, is involved in the encoding of light-tone pairings for future potential use, as it is the case of mediated learning processes. Specifically, our data provide an unforeseen role of the CaMKII-positive cells of the dHPC during associative memory between low-salience events. Overall, our results and previous studies^{1,11,13,14,52} clearly underlined the role of the hippocampus in sensory preconditioning being important in different phases such as the encoding of the associations between low salience stimuli such as lights and tones or enabling the value of the reinforcer to spread across different stimuli.

Other brain regions, such as the orbitofrontal, retrosplenial, and perirhinal cortices, have been linked to the processing of pairings between low-salience sensory cues^{6,8,10}. Interestingly, these regions anatomically interact with the hippocampus^{53–55}, suggesting unexplored brain circuits governed by the hippocampus allowing the brain to encode associations between different external environmental cues. In this sense, the identification of the dHPC, but not the vHPC, as a crucial brain region to encode these associations might give some novel clues to further understand the brain network underlying higher-order conditioning. A plausible brain circuit involved in the processing of low-salience stimuli could be neuronal projections between the retrosplenial cortex and the dHPC. In line with this, the inhibition of CaMKII-positive neurons from the dorsal hippocampus, which has been shown to project to the retrosplenial cortex⁵⁶, blocked the formation of mediated learning. Future work will be performed to tackle this hypothesis and to further decipher how brain networks control higher-order conditioning.

Overall, our results provide new insights on how hippocampal circuits control mediated learning formation by encoding low-salience stimuli pairings during the preconditioning phase of a $_{LT}$ SPC paradigm. It is relevant to fully understand how the brain allows the formation of mediated learning, which is a cognitive process that governs most of the animal and human daily decisions and could be involved in the basis of psychotic-like states^{57,58}.

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Additional information

Author Contributions

A.B-G. and J.P. contributed to the conception of the project and performed and analyzed all the experiments. C.R-D. help on all surgical procedures and immunohistochemistry assays. I.M-S. help with behavioral and immunohistochemistry assays. A.B-G. and J.P wrote the manuscript. All authors have corrected and revised the manuscript.

Additional files

Supplementary Information 

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Reviewer #1 (Public review):

Summary:

The study by Pinho et al. presents a novel behavioral paradigm for investigating higher-order conditioning in mice. The authors developed a task that creates associations between light and tone sensory cues, driving mediated learning. They observed sex differences in task acquisition, with females demonstrating faster-mediated learning compared to males. Using fiber photometry and chemogenetic tools, the study reveals that the dorsal hippocampus (dHPC) plays a central role in encoding mediated learning. These findings are crucial for understanding how environmental cues, which are not directly linked to positive/negative

outcomes, contribute to associative learning. Overall, the study is well-designed, with robust results, and the experimental approach aligns with the study's objectives.

Strengths:

- (1) The authors develop a robust behavioral paradigm to examine higher-order associative learning in mice.
- (2) They discover a sex-specific component influencing mediated learning, with females exhibiting enhanced learning abilities.
- (3) Using fiber photometry and chemogenetic techniques, the authors identify the dorsal hippocampus but not the ventral hippocampus, which plays a crucial role for encoding mediated learning.

Weaknesses:

- (1) The study would be strengthened by further elaboration on the rationale for investigating specific cell types within the hippocampus.
- (2) The analysis of photometry data could be improved by distinguishing between early and late responses, as well as enhancing the overall presentation of the data.
- (3) The manuscript would benefit from revisions to improve clarity and readability.

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Reviewer #2 (Public review):

Summary:

Pinho et al. developed a new auditory-visual sensory preconditioning procedure in mice and examined the contribution of the dorsal and ventral hippocampus to learning in this task. Using photometry they observed activation of the dorsal and ventral hippocampus during sensory preconditioning and conditioning. Finally, the authors combined their sensory preconditioning task with DREADDs to examine the effect of inhibiting specific cell populations (CaMKII and PV) in the DH on the formation and retrieval/expression of mediated learning.

Strengths:

The authors provide one of the first demonstrations of auditory-visual sensory preconditioning in male mice. Research on the neurobiology of sensory preconditioning has primarily used rats as subjects. The development of a robust protocol in mice will be beneficial to the field, allowing researchers to take advantage of the many transgenic mouse lines. Indeed, in this study, the authors take advantage of a PV-Cre mouse line to examine the role of hippocampal PV cells in sensory preconditioning.

Weaknesses:

- (1) The authors report that sensory preconditioning was observed in both male and female mice. However, their data only supports sensory preconditioning in male mice. In female mice, both paired and unpaired presentations of the light and tone in stage 1 led to increased freezing to the tone at test. In this case, fear to the tone could be attributed to factors other than sensory preconditioning, for example, generalization of fear between the auditory and visual stimulus.

(2) In the photometry experiment, the authors report an increase in neural activity in the hippocampus during both phase 1 (sensory preconditioning) and phase 2 (conditioning). In the subsequent experiment, they inhibit neural activity in the DH during phase 1 (sensory preconditioning) and the probe test, but do not include inhibition during phase 2 (conditioning). It was not clear why they didn't carry forward investigating the role of the hippocampus during phase 2 conditioning. Sensory preconditioning could occur due to the integration of the tone and shock during phase two, or retrieval and chaining of the tone-light-shock memories at test. These two possibilities cannot be differentiated based on the data. Given that we do not know at which stage the mediate learning is occurring, it would have been beneficial to additionally include inhibition of the DH during phase 2.

(3) In the final experiment, the authors report that inhibition of the dorsal hippocampus during the sensory preconditioning phase blocked mediated learning. While this may be the case, the failure to observe sensory preconditioning at test appears to be due more to an increase in baseline freezing (during the stimulus off period), rather than a decrease in freezing to the conditioned stimulus. Given the small effect, this study would benefit from an experiment validating that administration of J60 inhibited DH cells. Further, given that the authors did not observe any effect of DREADD inhibition in PV cells, it would also be important to validate successful cellular silencing in this protocol.

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Reviewer #3 (Public review):

Summary:

Pinho et al. investigated the role of the dorsal vs ventral hippocampus and the gender differences in mediated learning. While previous studies already established the engagement of the hippocampus in sensory preconditioning, the authors here took advantage of freely-moving fiber photometry recording and chemogenetics to observe and manipulate sub-regions of the hippocampus (dorsal vs. ventral) in a cell-specific manner. The authors first found sex differences in the preconditioning phase of a sensory preconditioning procedure, where males required more preconditioning training than females for mediating learning to manifest, and where females displayed evidence of mediated learning even when neutral stimuli were never presented together within the session.

After validation of a sensory preconditioning procedure in mice using light and tone neutral stimuli and a mild foot shock as the unconditioned stimulus, the authors used fiber photometry to record from all neurons vs. parvalbumin_positive_only neurons in the dorsal hippocampus or ventral hippocampus of male mice during both preconditioning and conditioning phases. They found increased activity of all neurons, as well as PV+_only neurons in both sub-regions of the hippocampus during both preconditioning and conditioning phases. Finally, the authors found that chemogenetic inhibition of CaMKII+ neurons in the dorsal, but not ventral, hippocampus specifically prevented the formation of an association between the two neutral stimuli (i.e., light and tone cues), but not the direct association between the light cue and the mild foot shock. This set of data: (1) validates the mediated learning in mice using a sensory preconditioning protocol, and stresses the importance of taking sex effect into account; (2) validates the recruitment of dorsal and ventral hippocampi during preconditioning and conditioning phases; and (3) further establishes the specific role of CaMKII+ neurons in the dorsal but not ventral hippocampus in the formation of an association between two neutral stimuli, but not between a neutral-stimulus and a mild foot shock.

Strengths:

The authors developed a sensory preconditioning procedure in mice to investigate mediated learning using light and tone cues as neutral stimuli, and a mild foot shock as the unconditioned stimulus. They provide evidence of a sex effect in the formation of light-cue association. The authors took advantage of fiber-photometry and chemogenetics to target sub-regions of the hippocampus, in a cell-specific manner and investigate their role during different phases of a sensory conditioning procedure.

Weaknesses:

The authors went further than previous studies by investigating the role of sub-regions of the hippocampus in mediated learning, however, there are several weaknesses that should be noted:

(1) This work first validates mediated learning in a sensory preconditioning procedure using light and tone cues as neutral stimuli and a mild foot shock as the unconditioned stimulus, in both males and females. They found interesting sex differences at the behavioral level, but then only focused on male mice when recording and manipulating the hippocampus. The authors do not address sex differences at the neural level.

(2) As expected in fear conditioning, the range of inter-individual differences is quite high. Mice that didn't develop a strong light→shock association, as evidenced by a lower percentage of freezing during the Probe Test Light phase, should manifest a low percentage of freezing during the Probe Test Tone phase. It would be interesting to test for a correlation between the level of freezing during mediated vs test phases.

(3) The use of a synapsin promoter to transfect neurons in a non-specific manner does not bring much information. The authors applied a more specific approach to target PV+ neurons only, and it would have been more informative to keep with this cell-specific approach, for example by looking also at somatostatin+ inter-neurons.

(4) The authors observed event-related Ca²⁺ transients on hippocampal pan-neurons and PV+ inter-neurons using fiber photometry. They then used chemogenetics to inhibit CaMKII+ hippocampal neurons, which does not logically follow. It does not undermine the main finding of CaMKII+ neurons of the dorsal, but not ventral, hippocampus being involved in the preconditioning, but not conditioning, phase. However, observing CaMKII+ neurons (using fiber photometry) in mice running the same task would be more informative, as it would indicate when these neurons are recruited during different phases of sensory preconditioning. Applying then optogenetics to cancel the observed event-related transients (e.g., during the presentation of light and tone cues, or during the foot shock presentation) would be more appropriate.

(5) Probe tests always start with the "Probe Test Tone", followed by the "Probe Test Light". "Probe Test Tone" consists of an extinction session, which could affect the freezing response during "Probe Test Light" (e.g., Polack et al. (<http://dx.doi.org/10.3758/s13420-013-0119-5>)). Preferably, adding a group of mice with a Probe Test Light with no Probe Test Tone could help clarify this potential issue. The authors should at least discuss the possibility that the tone extinction session prior to the "Probe Test Light" could have affected the freezing response to the light cue.

<https://doi.org/10.7554/eLife.105863.1.sa2>

Reviewer #4 (Public review):

Summary

Pinho et al use in vivo calcium imaging and chemogenetic approaches to examine the involvement of hippocampal sub-regions across the different stages of a sensory preconditioning task in mice. They find clear evidence for sensory preconditioning in male but not female mice. They also find that, in the male mice, CaMKII-positive neurons in the dorsal hippocampus: (1) encode the audio-visual association that forms in stage 1 of the task, and (2) retrieve/express sensory preconditioned fear to the auditory stimulus at test. These findings are supported by evidence that ranges from incomplete to convincing. They will be valuable to researchers in the field of learning and memory.

Abstract

Please note that sensory preconditioning doesn't require the stage 1 stimuli to be presented repeatedly or simultaneously.

"Finally, we combined our sensory preconditioning task with chemogenetic approaches to assess the role of these two hippocampal subregions in mediated learning." This implies some form of inhibition of hippocampal neurons in stage 2 of the protocol, as this is the only stage of the protocol that permits one to make statements about mediated learning. However, it is clear from what follows that the authors interrogate the involvement of hippocampal sub-regions in stages 1 and 3 of the protocol - not stage 2. As such, most statements about mediated learning throughout the paper are potentially misleading (see below for a further elaboration of this point). If the authors persist in using the term mediated learning to describe the response to a sensory preconditioned stimulus, they should clarify what they mean by mediated learning at some point in the introduction. Alternatively, they might consider using a different phrase such as "sensory preconditioned responding".

Introduction

"Low-salience" is used to describe stimuli such as tone, light, or odour that do not typically elicit responses that are of interest to experimenters. However, a tone, light, or odour can be very salient even though they don't elicit these particular responses. As such, it would be worth redescribing the "low-salience" stimuli in some other terms.

"These higher-order conditioning processes, also known as mediated learning, can be captured in laboratory settings through sensory preconditioning procedures^{2,6-11}." Higher-order conditioning and mediated learning are not interchangeable terms: e.g., some forms of second-order conditioning are not due to mediated learning. More generally, the use of mediated learning is not necessary for the story that the authors develop in the paper and could be replaced for accuracy and clarity. E.g., "These higher-order conditioning processes can be studied in the laboratory using sensory preconditioning procedures^{2,6-11}."

In reference to Experiment 2, it is stated that: "However, when light and tone were separated on time (Unpaired group), male mice were not able to exhibit mediated learning response (Figure 2B) whereas their response to the light (direct learning) was not affected (Figure 2D). On the other hand, female mice still present a lower but significant mediated learning response (Figure 2C) and normal direct learning (Figure 2E). Finally, in the No-Shock group, both male (Figure 2B and 2D) and female mice (Figure 2C and 2E) did not present either mediated or direct learning, which also confirmed that the exposure to the tone or light during Probe Tests do not elicit any behavioral change by themselves as the presence of the electric footshock is required to obtain a reliable mediated and direct learning responses." The absence of a difference between the paired and unpaired female mice should not be described as "significant mediated learning" in the latter. It should be taken to indicate that performance in the females is due to generalization between the tone and light. That is, there is no sensory preconditioning in the female mice. The description of performance in the No-shock group really shouldn't be in terms of mediated or direct learning: that is, this group is another control for assessing the presence of sensory preconditioning in the group of

interest. As a control, there is no potential for them to exhibit sensory preconditioning, so their performance should not be described in a way that suggests this potential.

Methods - Behavior

I appreciate the reasons for testing the animals in a new context. This does, however, raise other issues that complicate the interpretation of any hippocampal engagement: e.g., exposure to a novel context may engage the hippocampus for exploration/encoding of its features - hence, it is engaged for retrieving/expressing sensory preconditioned fear to the tone. This should be noted somewhere in the paper given that one of its aims is to shed light on the broader functioning of the hippocampus in associative processes.

This general issue - that the conditions of testing were such as to force engagement of the hippocampus - is amplified by two further features of testing with the tone. The first is the presence of background noise in the training context and its absence in the test context. The second is the fact that the tone was presented for 30 s in stage 1 and then continuously for 180s at test. Both changes could have contributed to the engagement of the hippocampus as they introduce the potential for discrimination between the tone that was trained and tested.

Results - Behavior

The suggestion of sex differences based on differences in the parameters needed to generate sensory preconditioning is interesting. Perhaps it could be supported through some set of formal analyses. That is, the data in supplementary materials may well show that the parameters needed to generate sensory preconditioning in males and females are not the same. However, there needs to be some form of statistical comparison to support this point. As part of this comparison, it would be neat if the authors included body weight as a covariate to determine whether any interactions with sex are moderated by body weight.

What is the value of the data shown in Figure 1 given that there are no controls for unpaired presentations of the sound and light? In the absence of these controls, the experiment cannot have shown that "Female and male mice show mediated learning using an auditory-visual sensory preconditioning task" as implied by its title. Minimally, this experiment should be relabelled.

"Altogether, this data confirmed that we successfully set up an LTSPC protocol in mice and that this behavioral paradigm can be used to further study the brain circuits involved in higher-order conditioning."

Please insert the qualifier that LTSPC was successfully established in male mice. There is no evidence of LTSPC in female mice.

Results - Brain

"Notably, the inhibition of CaMKII-positive neurons in the dHPC (i.e. J60 administration in DREADD-Gi mice) during preconditioning (Figure 4B), but not before the Probe Test 1 (Figure 4B), fully blocked mediated, but not direct learning (Figure 4D)."

The right panel of Figure 4B indicates no difference between the controls and Group DPC in the percent change in freezing from OFF to ON periods of the tone. How does this fit with the claim that CaMKII-positive neurons in the dorsal hippocampus regulate associative formation during the session of tone-light exposures in stage 1 of sensory preconditioning?

Discussion

"When low salience stimuli were presented separated on time or when the electric footshock was absent, mediated and direct learning were abolished in male mice. In female mice, although light and tone were presented separately during the preconditioning phase, mediated learning was reduced but still present, which implies that female mice are still able

to associate the two low-salience stimuli."

This doesn't quite follow from the results. The failure of the female unpaired mice to withhold their freezing to the tone should not be taken to indicate the formation of a light-tone association across the very long interval that was interpolated between these stimulus presentations. It could and should be taken to indicate that, in female mice, freezing conditioned to the light simply generalized to the tone (i.e., these mice could not discriminate well between the tone and light).

"Indeed, our data suggests that when hippocampal activity is modulated by the specific manipulation of hippocampal subregions, this brain region is not involved during retrieval." Does this relate to the results that are shown in the right panel of Figure 4B, where there is no significant difference between the different groups? If so, how does it fit with the results shown in the left panel of this figure, where differences between the groups are observed?

"In line with this, the inhibition of CaMKII-positive neurons from the dorsal hippocampus, which has been shown to project to the retrosplenial cortex⁵⁶, blocked the formation of mediated learning."

Is this a reference to the findings shown in Figure 4B and, if so, which of the panels exactly? That is, one panel appears to support the claim made here while the other doesn't. In general, what should the reader make of data showing the percent change in freezing from stimulus OFF to stimulus ON periods?

<https://doi.org/10.7554/eLife.105863.1.sa1>

Author response:

Reviewer #1 (Public review):

Summary:

The study by Pinho et al. presents a novel behavioral paradigm for investigating higher-order conditioning in mice. The authors developed a task that creates associations between light and tone sensory cues, driving mediated learning. They observed sex differences in task acquisition, with females demonstrating faster-mediated learning compared to males. Using fiber photometry and chemogenetic tools, the study reveals that the dorsal hippocampus (dHPC) plays a central role in encoding mediated learning. These findings are crucial for understanding how environmental cues, which are not directly linked to positive/negative outcomes, contribute to associative learning. Overall, the study is well-designed, with robust results, and the experimental approach aligns with the study's objectives.

Strengths:

(1) The authors develop a robust behavioral paradigm to examine higher-order associative learning in mice.

(2) They discover a sex-specific component influencing mediated learning, with females exhibiting enhanced learning abilities.

(3) Using fiber photometry and chemogenetic techniques, the authors identify the dorsal hippocampus but not the ventral hippocampus, which plays a crucial for encoding mediated learning.

Weaknesses:

(1) The study would be strengthened by further elaboration on the rationale for investigating specific cell types within the hippocampus.

We will add more information to better explain the rationale of our experiments and/or manipulations.

(2) The analysis of photometry data could be improved by distinguishing between early and late responses, as well as enhancing the overall presentation of the data.

We will provide new photometry analysis to differentiate between early and late responses during stimuli presentations.

(3) The manuscript would benefit from revisions to improve clarity and readability.

We will improve the clarity and readability of our manuscript.

Reviewer #2 (Public review):

Summary:

Pinho et al. developed a new auditory-visual sensory preconditioning procedure in mice and examined the contribution of the dorsal and ventral hippocampus to learning in this task. Using photometry they observed activation of the dorsal and ventral hippocampus during sensory preconditioning and conditioning. Finally, the authors combined their sensory preconditioning task with DREADDs to examine the effect of inhibiting specific cell populations (CaMKII and PV) in the DH on the formation and retrieval/expression of mediated learning.

Strengths:

The authors provide one of the first demonstrations of auditory-visual sensory preconditioning in male mice. Research on the neurobiology of sensory preconditioning has primarily used rats as subjects. The development of a robust protocol in mice will be beneficial to the field, allowing researchers to take advantage of the many transgenic mouse lines. Indeed, in this study, the authors take advantage of a PV-Cre mouse line to examine the role of hippocampal PV cells in sensory preconditioning.

Weaknesses:

(1) The authors report that sensory preconditioning was observed in both male and female mice. However, their data only supports sensory preconditioning in male mice. In female mice, both paired and unpaired presentations of the light and tone in stage 1 led to increased freezing to the tone at test. In this case, fear to the tone could be attributed to factors other than sensory preconditioning, for example, generalization of fear between the auditory and visual stimulus.

To address the pertinent doubt raised by the reviewer, we will perform new experiments to generate a new unpaired group in female mice through the increase of the temporal interval between light and tone exposure during the preconditioning phase. We believe this new results will bring additional information to better understand the performance of female mice in sensory preconditioning.

(2) In the photometry experiment, the authors report an increase in neural activity in the hippocampus during both phase 1 (sensory preconditioning) and phase 2 (conditioning). In the subsequent experiment, they inhibit neural activity in the DH during phase 1

(sensory preconditioning) and the probe test, but do not include inhibition during phase 2 (conditioning). It was not clear why they didn't carry forward investigating the role of the hippocampus during phase 2 conditioning. Sensory preconditioning could occur due to the integration of the tone and shock during phase two, or retrieval and chaining of the tone-light-shock memories at test. These two possibilities cannot be differentiated based on the data. Given that we do not know at which stage the mediate learning is occurring, it would have been beneficial to additionally include inhibition of the DH during phase 2.

We will perform new experiments to generate novel data by inhibiting the CamK-positive neurons of the dorsal hippocampus during the conditioning phase.

(3) In the final experiment, the authors report that inhibition of the dorsal hippocampus during the sensory preconditioning phase blocked mediated learning. While this may be the case, the failure to observe sensory preconditioning at test appears to be due more to an increase in baseline freezing (during the stimulus off period), rather than a decrease in freezing to the conditioned stimulus. Given the small effect, this study would benefit from an experiment validating that administration of J60 inhibited DH cells. Further, given that the authors did not observe any effect of DREADD inhibition in PV cells, it would also be important to validate successful cellular silencing in this protocol.

By combining chemogenetic and fiber photometry approaches, we will perform a control experiments to demonstrate that our chemogenetic experiments are decreasing CAMK- or PV-dependent activity in dorsal and ventral hippocampus.

Reviewer #3 (Public review):

Summary:

Pinho et al. investigated the role of the dorsal vs ventral hippocampus and the gender differences in mediated learning. While previous studies already established the engagement of the hippocampus in sensory preconditioning, the authors here took advantage of freely-moving fiber photometry recording and chemogenetics to observe and manipulate sub-regions of the hippocampus (dorsal vs. ventral) in a cell-specific manner. The authors first found sex differences in the preconditioning phase of a sensory preconditioning procedure, where males required more preconditioning training than females for mediating learning to manifest, and where females displayed evidence of mediated learning even when neutral stimuli were never presented together within the session.

After validation of a sensory preconditioning procedure in mice using light and tone neutral stimuli and a mild foot shock as the unconditioned stimulus, the authors used fiber photometry to record from all neurons vs. parvalbumin_positive_only neurons in the dorsal hippocampus or ventral hippocampus of male mice during both preconditioning and conditioning phases. They found increased activity of all neurons, as well as PV+ only neurons in both sub-regions of the hippocampus during both preconditioning and conditioning phases. Finally, the authors found that chemogenetic inhibition of CaMKII+ neurons in the dorsal, but not ventral, hippocampus specifically prevented the formation of an association between the two neutral stimuli (i.e., light and tone cues), but not the direct association between the light cue and the mild foot shock. This set of data: (1) validates the mediated learning in mice using a sensory preconditioning protocol, and stresses the importance of taking sex effect into account; (2) validates the recruitment of dorsal and ventral hippocampi during preconditioning and conditioning phases; and (3) further establishes the specific role of CaMKII+ neurons

in the dorsal but not ventral hippocampus in the formation of an association between two neutral stimuli, but not between a neutral-stimulus and a mild foot shock.

Strengths:

The authors developed a sensory preconditioning procedure in mice to investigate mediated learning using light and tone cues as neutral stimuli, and a mild foot shock as the unconditioned stimulus. They provide evidence of a sex effect in the formation of light-cue association. The authors took advantage of fiber-photometry and chemogenetics to target sub-regions of the hippocampus, in a cell-specific manner and investigate their role during different phases of a sensory conditioning procedure.

Weaknesses:

The authors went further than previous studies by investigating the role of sub-regions of the hippocampus in mediated learning, however, there are several weaknesses that should be noted:

(1) This work first validates mediated learning in a sensory preconditioning procedure using light and tone cues as neutral stimuli and a mild foot shock as the unconditioned stimulus, in both males and females. They found interesting sex differences at the behavioral level, but then only focused on male mice when recording and manipulating the hippocampus. The authors do not address sex differences at the neural level.

As discussed above, we will perform additional experiment to evaluate the presence of a reliable sensory preconditioning in female mice. In addition, although observing sex differences at the neural level can be very interesting, we think that it is out of the scope of the present work. However, we will mention this issue/limitation in the Discussion in the new version of the manuscript.

(2) As expected in fear conditioning, the range of inter-individual differences is quite high. Mice that didn't develop a strong light-->shock association, as evidenced by a lower percentage of freezing during the Probe Test Light phase, should manifest a low percentage of freezing during the Probe Test Tone phase. It would interesting to test for a correlation between the level of freezing during mediated vs test phases.

We will provide correlations between the behavioral responses in both probe tests.

(3) The use of a synapsin promoter to transfect neurons in a non-specific manner does not bring much information. The authors applied a more specific approach to target PV+ neurons only, and it would have been more informative to keep with this cell-specific approach, for example by looking also at somatostatin+ inter-neurons.

We will better justify the use of specific promoters and the targeting of PV-positive neurons. We will also add discussion on potential interesting future experiments such as the targeting of other GABAergic subtypes.

(4) The authors observed event-related Ca²⁺ transients on hippocampal pan-neurons and PV+ inter-neurons using fiber photometry. They then used chemogenetics to inhibit CaMKII+ hippocampal neurons, which does not logically follow. It does not undermine the main finding of CaMKII+ neurons of the dorsal, but not ventral, hippocampus being involved in the preconditioning, but not conditioning, phase. However, observing CaMKII+ neurons (using fiber photometry) in mice running the same task would be more informative, as it would indicate when these neurons are recruited during different phases of sensory preconditioning. Applying then optogenetics to cancel the observed

event-related transients (e.g., during the presentation of light and tone cues, or during the foot shock presentation) would be more appropriate.

We will perform new experiments to analyze the activity of CAMK-positive neurons during light-tone associations during the preconditioning phase in male mice.

(5) Probe tests always start with the "Probe Test Tone", followed by the "Probe Test Light". "Probe Test Tone" consists of an extinction session, which could affect the freezing response during "Probe Test Light" (e.g., Polack et al. (<http://dx.doi.org/10.3758/s13420-013-0119-5>)). Preferably, adding a group of mice with a Probe Test Light with no Probe Test Tone could help clarify this potential issue. The authors should at least discuss the possibility that the tone extinction session prior to the "Probe Test Light" could have affected the freezing response to the light cue.

We will add discussion on this issue raised by the reviewer.

Reviewer #4 (Public review):

Summary

Pinho et al use in vivo calcium imaging and chemogenetic approaches to examine the involvement of hippocampal sub-regions across the different stages of a sensory preconditioning task in mice. They find clear evidence for sensory preconditioning in male but not female mice. They also find that, in the male mice, CaMKII-positive neurons in the dorsal hippocampus: (1) encode the audio-visual association that forms in stage 1 of the task, and (2) retrieve/express sensory preconditioned fear to the auditory stimulus at test. These findings are supported by evidence that ranges from incomplete to convincing. They will be valuable to researchers in the field of learning and memory.

Abstract

Please note that sensory preconditioning doesn't require the stage 1 stimuli to be presented repeatedly or simultaneously.

We will correct this wrong sentence in the abstract.

"Finally, we combined our sensory preconditioning task with chemogenetic approaches to assess the role of these two hippocampal subregions in mediated learning."

This implies some form of inhibition of hippocampal neurons in stage 2 of the protocol, as this is the only stage of the protocol that permits one to make statements about mediated learning. However, it is clear from what follows that the authors interrogate the involvement of hippocampal sub-regions in stages 1 and 3 of the protocol - not stage 2. As such, most statements about mediated learning throughout the paper are potentially misleading (see below for a further elaboration of this point). If the authors persist in using the term mediated learning to describe the response to a sensory preconditioned stimulus, they should clarify what they mean by mediated learning at some point in the introduction. Alternatively, they might consider using a different phrase such as "sensory preconditioned responding".

Through the text, we will avoid the term "mediated learning" and we will replace it with more accurate terms. In addition, we will interrogate the role of dHPC in Stage 2 as commented above.

Introduction

"Low-salience" is used to describe stimuli such as tone, light, or odour that do not typically elicit responses that are of interest to experimenters. However, a tone, light, or odour can be very salient even though they don't elicit these particular responses. As such, it would be worth redescribing the "low-salience" stimuli in some other terms.

We will substitute "low-salience" for "innocuous".

"These higher-order conditioning processes, also known as mediated learning, can be captured in laboratory settings through sensory preconditioning procedures^{2,6-11}."

Higher-order conditioning and mediated learning are not interchangeable terms: e.g., some forms of second-order conditioning are not due to mediated learning. More generally, the use of mediated learning is not necessary for the story that the authors develop in the paper and could be replaced for accuracy and clarity. E.g., "These higher-order conditioning processes can be studied in the laboratory using sensory preconditioning procedures^{2,6-11}."

Through the text, we will avoid the term "mediated learning" and we will replace it with more accurate terms.

In reference to Experiment 2, it is stated that: "However, when light and tone were separated on time (Unpaired group), male mice were not able to exhibit mediated learning response (Figure 2B) whereas their response to the light (direct learning) was not affected (Figure 2D). On the other hand, female mice still present a lower but significant mediated learning response (Figure 2C) and normal direct learning (Figure 2E). Finally, in the No-Shock group, both male (Figure 2B and 2D) and female mice (Figure 2C and 2E) did not present either mediated or direct learning, which also confirmed that the exposure to the tone or light during Probe Tests do not elicit any behavioral change by themselves as the presence of the electric footshock is required to obtain a reliable mediated and direct learning responses."

The absence of a difference between the paired and unpaired female mice should not be described as "significant mediated learning" in the latter. It should be taken to indicate that performance in the females is due to generalization between the tone and light. That is, there is no sensory preconditioning in the female mice. The description of performance in the No-shock group really shouldn't be in terms of mediated or direct learning: that is, this group is another control for assessing the presence of sensory preconditioning in the group of interest. As a control, there is no potential for them to exhibit sensory preconditioning, so their performance should not be described in a way that suggests this potential.

We will re-write the text to clarify the right comments raised by the Reviewer.

Methods - Behavior

I appreciate the reasons for testing the animals in a new context. This does, however, raise other issues that complicate the interpretation of any hippocampal engagement: e.g., exposure to a novel context may engage the hippocampus for exploration/encoding of its features - hence, it is engaged for retrieving/expressing sensory preconditioned fear to the tone. This should be noted somewhere in the paper given that one of its aims is to shed light on the broader functioning of the hippocampus in associative processes.

We will further discuss this aspect on the manuscript.

This general issue - that the conditions of testing were such as to force engagement of the hippocampus - is amplified by two further features of testing with the tone. The first is the presence of background noise in the training context and its absence in the test context. The second is the fact that the tone was presented for 30 s in stage 1 and then continuously for 180s at test. Both changes could have contributed to the engagement of the hippocampus as they introduce the potential for discrimination between the tone that was trained and tested.

We will consider the aspect raised by the reviewer on the manuscript.

Results - Behavior

The suggestion of sex differences based on differences in the parameters needed to generate sensory preconditioning is interesting. Perhaps it could be supported through some set of formal analyses. That is, the data in supplementary materials may well show that the parameters needed to generate sensory preconditioning in males and females are not the same. However, there needs to be some form of statistical comparison to support this point. As part of this comparison, it would be neat if the authors included body weight as a covariate to determine whether any interactions with sex are moderated by body weight.

We will add statistical comparisons between male and female mice.

What is the value of the data shown in Figure 1 given that there are no controls for unpaired presentations of the sound and light? In the absence of these controls, the experiment cannot have shown that "Female and male mice show mediated learning using an auditory-visual sensory preconditioning task" as implied by its title. Minimally, this experiment should be relabelled.

We will relabel Figure 1.

"Altogether, this data confirmed that we successfully set up an LTSPC protocol in mice and that this behavioral paradigm can be used to further study the brain circuits involved in higher-order conditioning."

Please insert the qualifier that LTSPC was successfully established in male mice. There is no evidence of LTSPC in female mice.

We will generate new experiments to try to demonstrate that SPC can be also observed in female mice.

Results - Brain

"Notably, the inhibition of CaMKII-positive neurons in the dHPC (i.e. J60 administration in DREADD-Gi mice) during preconditioning (Figure 4B), but not before the Probe Test 1 (Figure 4B), fully blocked mediated, but not direct learning (Figure 4D)."

The right panel of Figure 4B indicates no difference between the controls and Group DPC in the percent change in freezing from OFF to ON periods of the tone. How does this fit with the claim that CaMKII-positive neurons in the dorsal hippocampus regulate associative formation during the session of tone-light exposures in stage 1 of sensory preconditioning?

We will rephrase and add more Discussion regarding this section of the results to stick to what the graphs are showing. We will clarify that the group where dHPC activity is inhibited during preconditioning is the only one where the % of change is not significantly different from 0 (compared to the control or the group where the dHPC activity was modulated during the test).

Discussion

"When low salience stimuli were presented separated on time or when the electric footshock was absent, mediated and direct learning were abolished in male mice. In female mice, although light and tone were presented separately during the preconditioning phase, mediated learning was reduced but still present, which implies that female mice are still able to associate the two low-salience stimuli."

This doesn't quite follow from the results. The failure of the female unpaired mice to withhold their freezing to the tone should not be taken to indicate the formation of a light-tone association across the very long interval that was interpolated between these stimulus presentations. It could and should be taken to indicate that, in female mice, freezing conditioned to the light simply generalized to the tone (i.e., these mice could not discriminate well between the tone and light).

We will rewrite this part depending on the results observed in female mice.

"Indeed, our data suggests that when hippocampal activity is modulated by the specific manipulation of hippocampal subregions, this brain region is not involved during retrieval."

Does this relate to the results that are shown in the right panel of Figure 4B, where there is no significant difference between the different groups? If so, how does it fit with the results shown in the left panel of this figure, where differences between the groups are observed?

We will re-write it to clearly describe our results and we will also revise all the statistical analysis.

"In line with this, the inhibition of CaMKII-positive neurons from the dorsal hippocampus, which has been shown to project to the retrosplenial cortex56, blocked the formation of mediated learning."

Is this a reference to the findings shown in Figure 4B and, if so, which of the panels exactly? That is, one panel appears to support the claim made here while the other doesn't. In general, what should the reader make of data showing the percent change in freezing from stimulus OFF to stimulus ON periods?

We will rewrite the text to clearly describe our results, and we will also revise all the statistical analysis. In addition, we will better explain the data showing the % of change.

<https://doi.org/10.7554/eLife.105863.1.sa0>